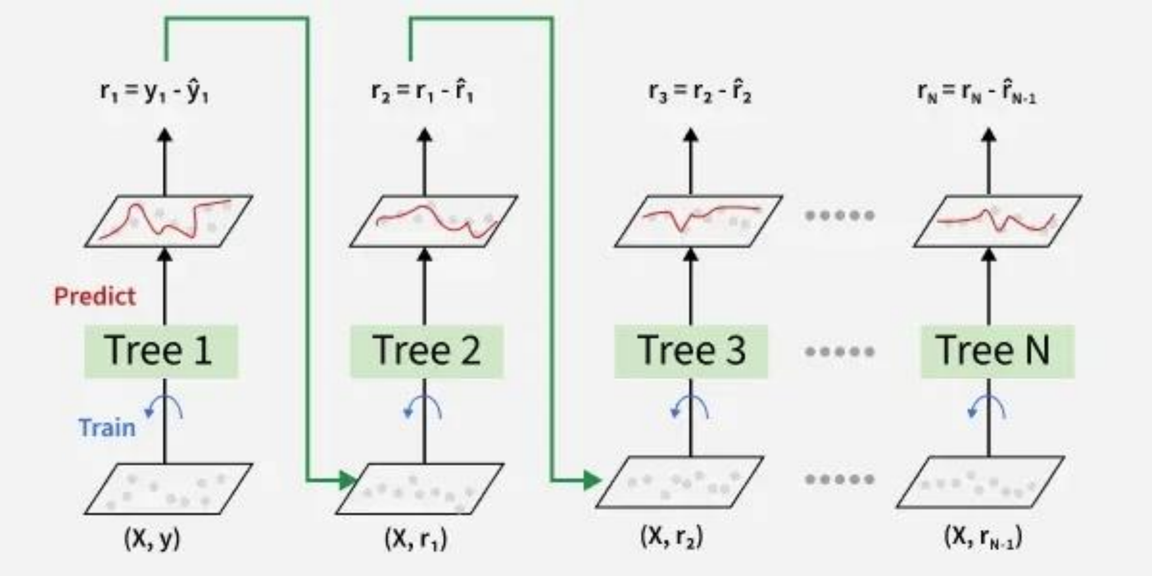


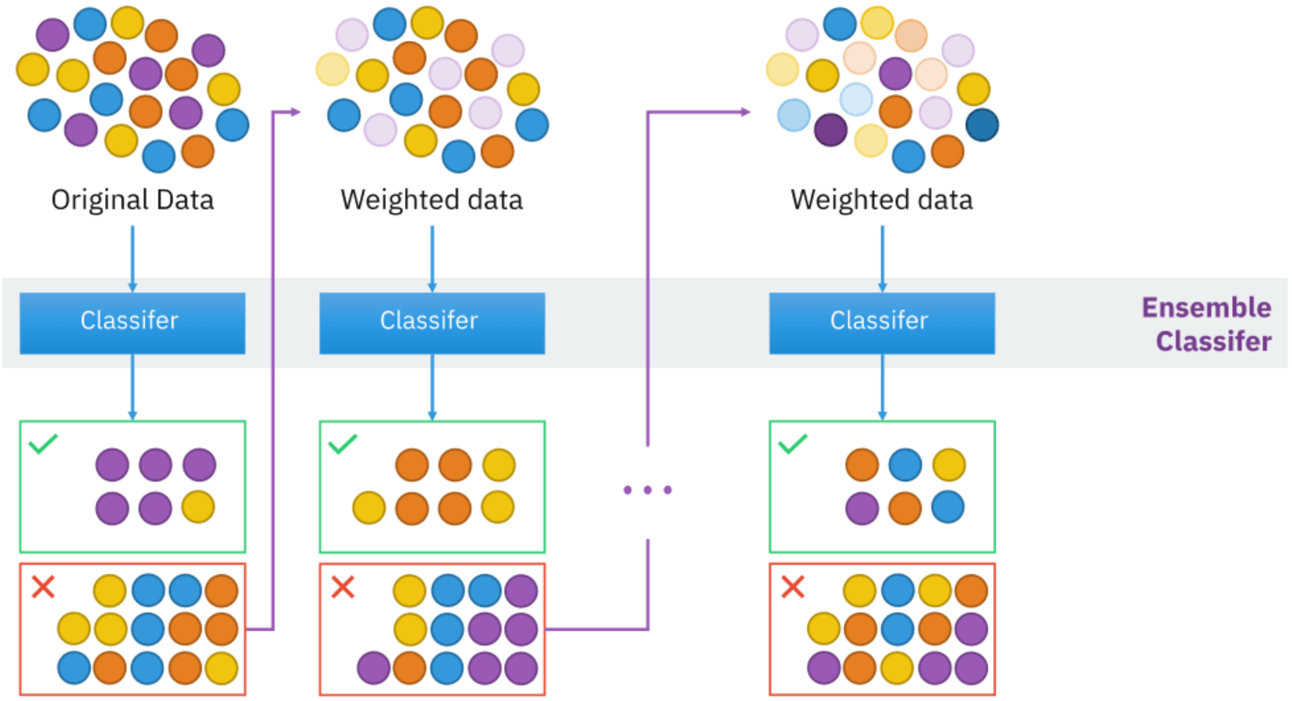
Gradient Boosting Foundations

Learning from Mistakes to Achieve Perfection

Instructor: Zaryab Ahmad Khan

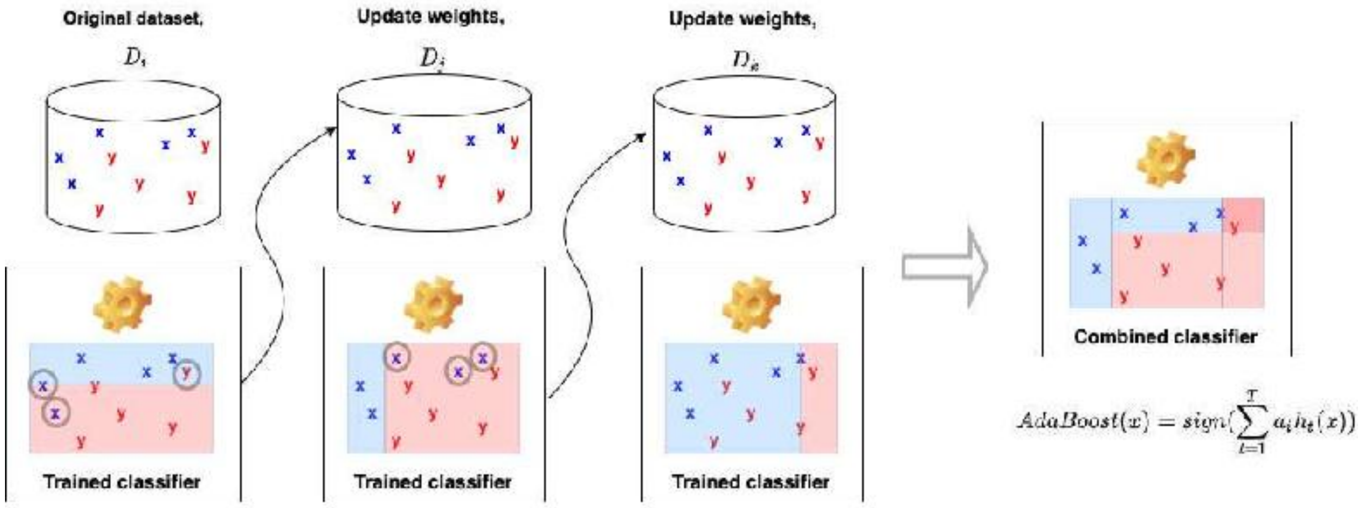
The Concept of Boosting

- **Sequential Learning**: Unlike Random Forest (where trees are built simultaneously), Boosting builds models one after another.
- **The Logic**: Each new model specifically tries to fix the mistakes made by the previous model.
- **Analogy**: It is like a student who takes a quiz, sees which questions they got wrong, and spends 100% of their time studying *only* those wrong answers for the next quiz.



AdaBoost (Adaptive Boosting)

- **Definition**: The classic boosting algorithm, widely used for basic classification tasks.
- **How it Works**: It gives more "weight" to the data points that were misclassified by the previous model, forcing the next model to focus on the hardest cases.



Gradient Boosting Regressor/Classifier

- **Definition**: A more advanced version of boosting that uses calculus (gradients) to minimize errors.
- **The Power**: It provides highly accurate predictions for standard tabular data (Excel-style data).
- **Efficiency**: It is the "gold standard" for winning many machine learning competitions because of its extreme accuracy